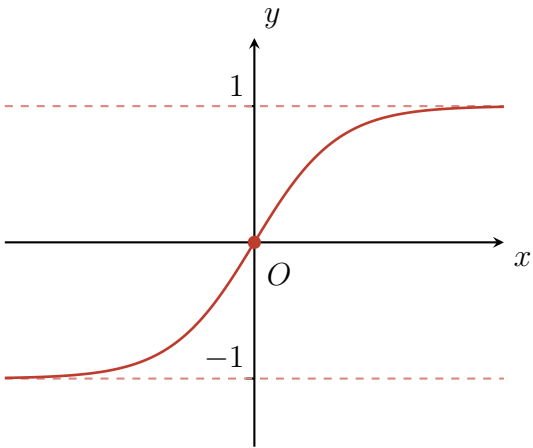
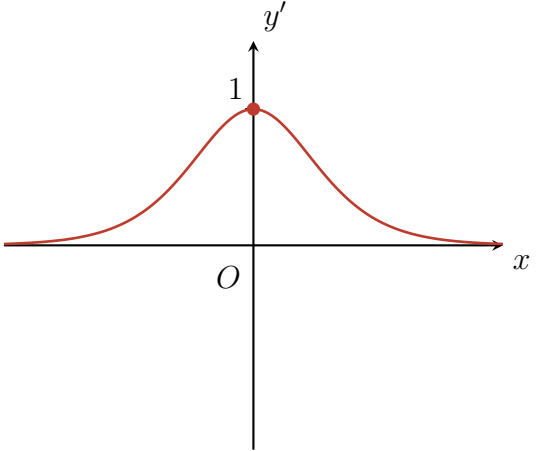


$$f(x) = \tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$



$$f'(x) = 1 - \tanh^2(x)$$